Methodology 1 — Final Experimental Analysis

Project: DoS Attack Analysis in SDN-Based VANETs

Scope: DoS injection, impact measurement, and static SDN source-based UDP filtering.

1. Methodology 1 Experiment Description

Methodology 1 evaluates the effect of a controlled Denial-of-Service (DoS) traffic load on an SDN-based VANET environment and verifies a basic mitigation mechanism. The experiment uses a Mininet-WiFi topology with two access points connected through an Open vSwitch SDN switch and a Ryu controller. SUMO was used separately to generate and collect vehicle mobility data. All attack traffic was confined to the local simulation/emulation environment.

The experiment was performed in three main stages. First, normal ICMP and UDP traffic was measured to establish a baseline. Second, controlled UDP traffic was generated from the attacker station at approximately 10 Mbps while legitimate traffic was observed, allowing the impact of the attack on latency to be measured. Third, an OpenFlow rule was installed on the attacker-side AP to drop IPv4 UDP packets originating from the attacker address (10.0.0.3). The attack was then repeated and the OpenFlow packet counter was examined to verify that the defense actively filtered the traffic. Legitimate UDP traffic was also tested while the defense was enabled.

2. Final Results Table

Metric	Normal	DoS	DoS + Defense
ICMP PDR	100%	100%	—
Average ICMP latency	0.792 ms	68.305 ms	—
UDP throughput	2.10 Mbps	10.5 Mbps	2.10 Mbps legitimate
Attacker packets dropped	—	—	18,036
Attacker bytes dropped	—	—	27,270,432 bytes
Approx. dropped data	—	—	26.01 MB

3. Final Metric Calculations

Packet Delivery Ratio: PDR = (received packets / sent packets) × 100. The baseline ICMP test received 20 of 20 packets, giving 100% PDR. The ICMP test during the DoS also received 20 of 20 packets, giving 100% PDR. The attack therefore manifested primarily as latency degradation rather than packet loss in this experiment.

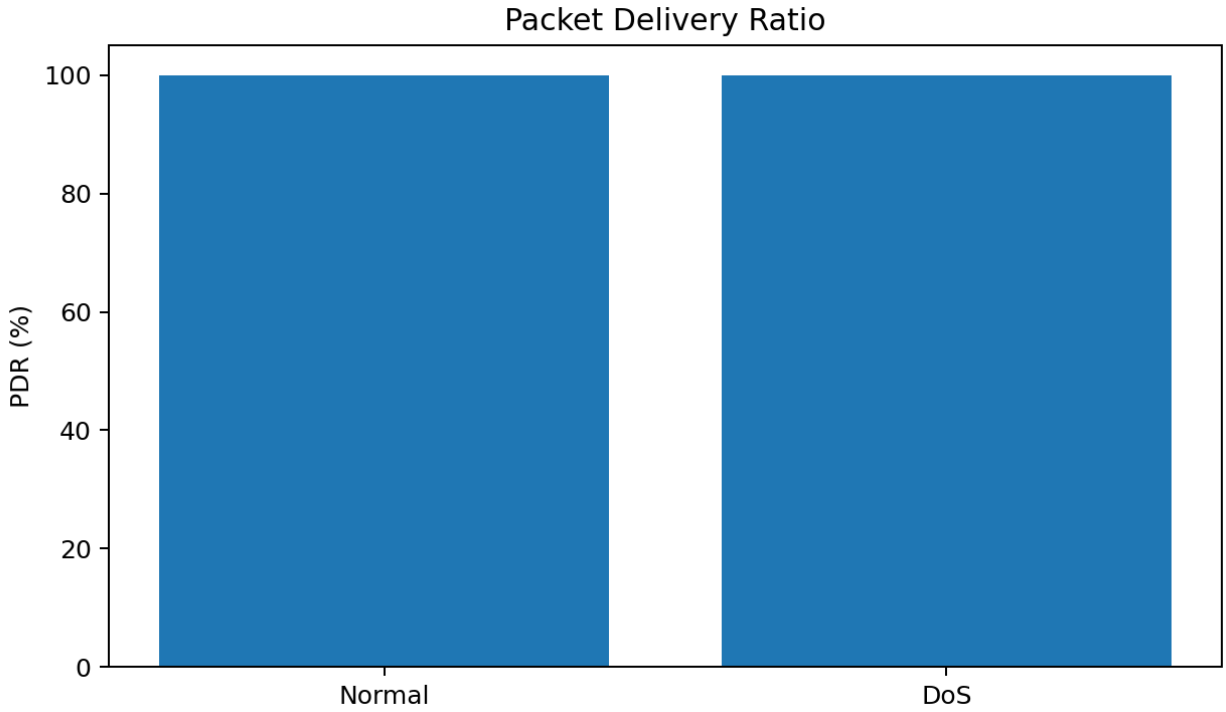
Latency increase: 68.305 – 0.792 = 67.513 ms. Percentage increase = ((68.305 – 0.792) / 0.792) × 100 ≈ **8524.4%**. The average latency during the attack was approximately **86.24×** the baseline latency.

Defense filtering: The AP2 OpenFlow defense rule recorded 18,036 dropped packets and 27,270,432 dropped bytes. This is approximately **26.01 MB** of filtered attacker traffic.

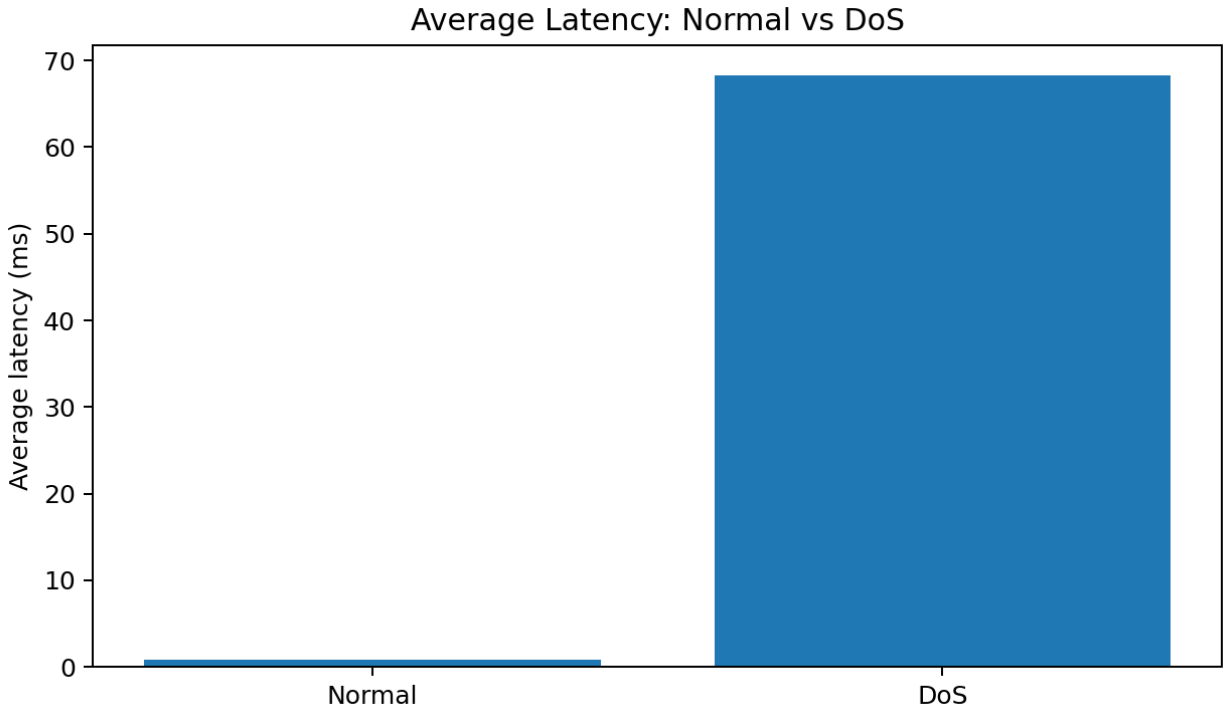
Legitimate traffic preservation: Legitimate UDP traffic measured 2.10 Mbps both in the normal test and during the defense test, indicating approximately 0% measured throughput degradation for that test.

4. Graphs

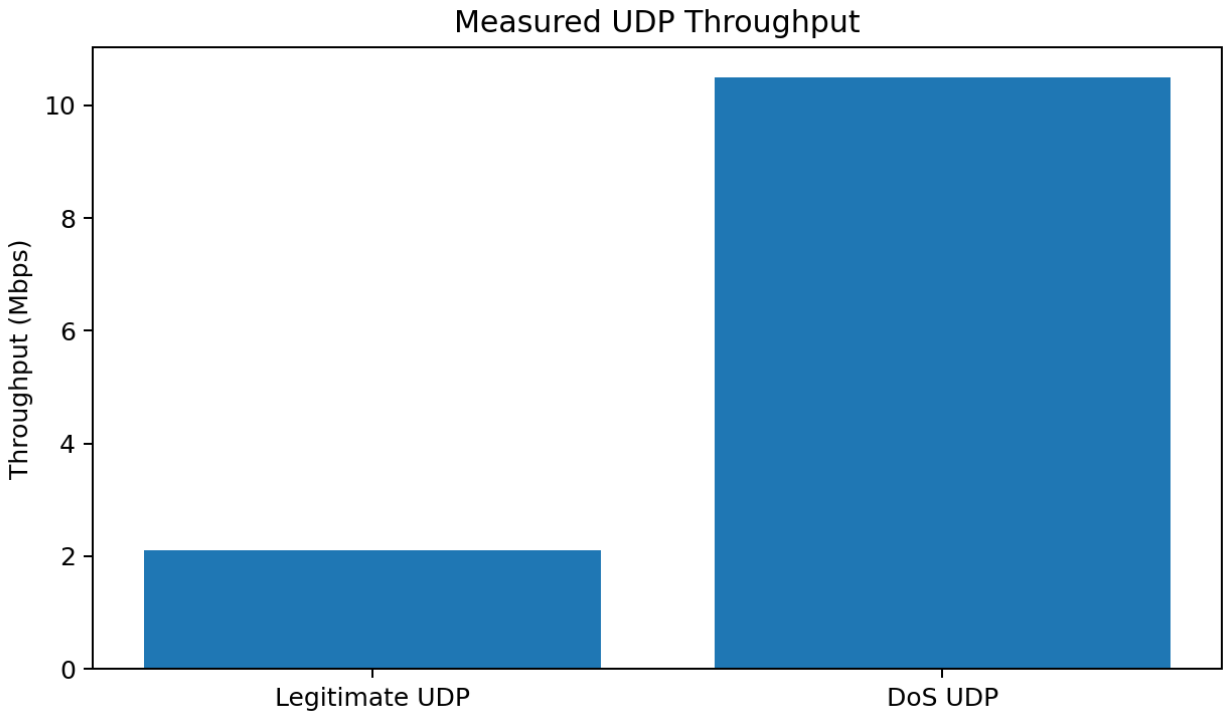
PDR



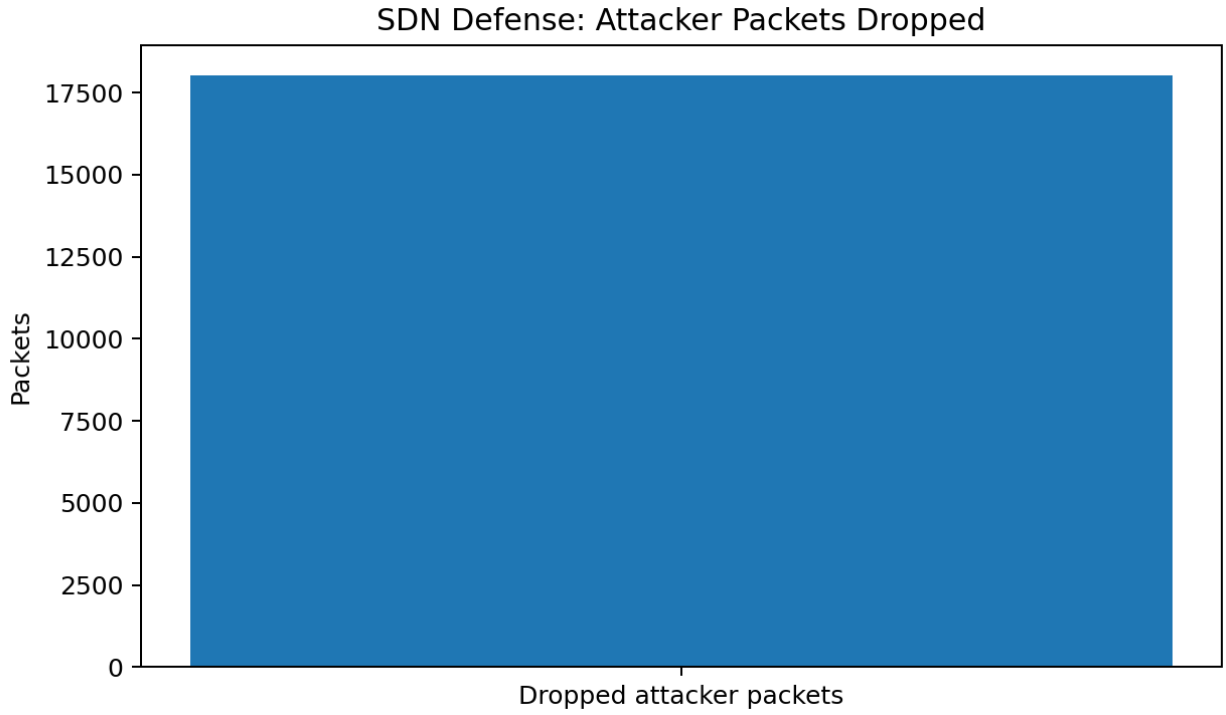
Latency



Throughput



Defense Packet Drops



5. Results and Discussion

The baseline experiment showed reliable communication between the legitimate stations, with 100% ICMP packet delivery and an average latency of 0.792 ms. Legitimate UDP traffic achieved 2.10 Mbps. When the controlled UDP DoS traffic was introduced, the measured ICMP packet delivery remained at 100%, but average latency increased sharply to 68.305 ms. This demonstrates that, in this particular topology and load, the main observable effect of the DoS traffic was increased communication delay rather than packet loss.

The static SDN defense successfully filtered the attacker traffic. The OpenFlow rule matching IPv4 UDP traffic from 10.0.0.3 recorded 18,036 dropped packets and 27,270,432 dropped bytes. At the same time, a legitimate UDP flow from sta2 to sta1 achieved 2.10 Mbps while the defense remained active. These observations indicate that the source-specific filtering rule blocked the identified attack traffic while allowing the tested legitimate flow to continue.

The current defense should be described as static source-based UDP filtering, not as a general rate limiter. A true rate-limiting implementation would require an OpenFlow meter or another explicit traffic-rate control mechanism. The results therefore establish the Methodology 1 baseline, attack impact, and basic SDN filtering capability without overstating the implemented mechanism.

6. Conclusion for Methodology 1

Methodology 1 has been experimentally completed. The measured results provide a baseline, a controlled DoS impact case, and a verified SDN filtering case. The experiment demonstrates substantial latency degradation under attack and confirms that the Ryu/OpenFlow defense can identify and drop UDP traffic from the configured attacker source while preserving the tested legitimate UDP communication.